

ARCHITECTURE DECISION FLOW — ADR ENGINE

Explicit evaluation criteria: integrity, failure behavior, security, performance, economics, evolution and operational proof.

ARCHITECTURE CONTROL POINTS

- The selected direction should be defensible independently of any single technology vendor.
- Rejected options are retained because their failure modes explain why the final architecture contains additional complexity.

PROBLEM

Context

business need + current state
system boundary

Constraints

NFRs • legacy coupling
team / delivery / platform constraints

bound

OPTIONS

Option A

application-centric / synchronous
local state + provider SDK

Option C

fully event-driven platform
maximum decoupling / operational complexity

Option B

reusable platform runtime
control/data plane separation

EVALUATION MATRIX

Performance

hot-path latency
elasticity • tail behavior

Security

tenant isolation
least privilege • auditability

Failure Behavior

blast radius • retry safety
recovery semantics

Evolution

provider/framework replacement
platform reuse

Architecture Integrity

state ownership • authority
contract clarity

Economics

route choice • cache
usage attribution

DECISION & CONSEQUENCE

Selected ADR

decision • rationale
trade-offs • residual risk

Implementation Consequence

interfaces • runtime semantics
operational ownership

Validation

production telemetry
failure / security / load evidence

translate

prove

